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Certificateless Sequential Aggregate Signature Scheme on NTRU Lattice

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Abstract — Having the advantages of certificateless signature and the aggregate signature at the same time, certificateless aggregate signature has been widely applied in e-business, e-government and software security since it was proposed in 2007. Although a number of certificateless aggregate signature schemes have been proposed, all of them are based on the classic number theory problem, which are no longer secure in the quantum era. In this paper, a certificateless sequential aggregate signature over number theory research unit lattice is proposed, which is proven to be secure in random oracle model. Moreover, we extend the new scheme into an efficient certificate-based sequential aggregate signature which is also secure in quantum era.

Key words — Certificateless, Sequential, Aggregate, Signature, Number theory research unit, Secure, Lattice.

I. Introduction

Since it was proposed in 1976, the digital signature scheme has more than 40 years' history. So constructing appropriately secure digital signature schemes or implementing those schemes is no longer the research emphasis. With the advent of lightweight cryptography, the main difficulty today in digital signature is the lightweight authentication.

On the one hand, in the traditional Public key infrastructure (PKI), a trusted Certificate authority (CA) composes certificates to ensure the authenticity of the users. This brings the vexing problem—certificate management problem. In order to deal with it, Identity-based Public key cryptography (IB-PKC) was first proposed in Ref.[1]. In the IB-PKC, the public key is just the product of the user's identity when the private key is generated by the trusted Private key generator (PKG) and the user's identity. It is obvious that the IB-PKC has the advantage of certificateless. However, it suffers from the key escrow problem. Generally speaking, the PKG knows all users' private keys. To overcome it, Certificateless Public key cryptography (CL-PKC) was proposed in Ref.[2], which has the significant advantages of certificateless and no escrow feature.

On the other hand, in order to compress the signatures on many different messages generated by many different users, Boneh *et al.* introduced the definition of Aggregate signature (AS) in Ref.[3]. The aggregate signature provides non-repudiation simultaneously on message m_1 for user 1, message m_2 for user 2, and so on. However, such non-repudiation just holds for each user regardless of whether other users are malicious. In order to solve this problem, Lysyanskaya *et al.* proposed the syntax of Sequential aggregate signature (SAS) in Ref.[4]. In a sequential aggregate signature scheme, the aggregation for certificate chains should be performed incrementally and sequentially.

In 2007, Gong *et al.* firstly introduced the definition of Certificateless aggregate signature (CLAS) in Ref.[5]. Since it was proposed, there have been several certificateless aggregate signature schemes, such as the schemes in Refs.[5-9]. However, all of them are based on the classic number theory and Shor has indicated that they would no longer be secure in quantum era[10].

In recent years, lattice-based cryptography has attracted significant interest. And over more than 20

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years' development, many beautiful constructions on lattice continued to spring up, such as, lattice-based schemes[11-28]. In view of the recent progress about quantum computers, looking for a quantum-immune Certificateless SAS (CLSAS) is very urgent. Considering the advantages of lattice-based cryptography, lattice-based certificateless sequential aggregate signature may be a good attempt.

In this paper, the first CLSAS scheme on N-TRU(Number Theory Research Unit) lattice is proposed. We prove it is existentially unforgeable against strong adversaries in the random oracle model when the Small integer solution (SIS) problem on NTRU lattice is hard. Based on the new CLSAS scheme, a Certificate-based SAS (CBSAS) scheme is also proposed. The security proof is similar to that of the new CLSAS scheme.

II. Preliminaries

1. Notation

Throughout this paper, the security parameter $n = 2^t$ is a positive integer where the integer t is larger than 2. $\mathbb{R}$ and $\mathbb{Z}$ are the real space and integer space, respectively. We will work in the ring $R = \mathbb{Z}[x]/(x^n+1)$ and ring $R_q = \mathbb{Z}_q[x]/(x^n+1)$ where the prime q is bigger than 5. It is also satisfied that x^n+1 can split into k_q irreducible factors modulo prime q, and k_q is equal to 2 or n. $R^\times$ denotes the set of invertible elements in R. If the vector x is selected from $\mathbb{R}^n$, then $||x||$ denotes the Euclidean norm of x. And for a matrix $\boldsymbol{A}$, let $\boldsymbol{a}_i$ be the ith column of $\boldsymbol{A}$ and $||\boldsymbol{A}||$ are defined as $max_i(||\boldsymbol{a}_i||)$.

Let $f = \sum_{i=0}^{n-1} f_i x^i$ and $g = \sum_{i=0}^{n-1} g_i x^i$ be polynomials in R.

- fg denotes polynomial multiplication in R, while $f * g = fg \bmod (x^n+1)$. Where $f * g$ denotes Modular multiplication.

- (f) is the vector whose coordinates are respectively $f_0, ..., f_{n-1}$. $(f, g) \in \mathbb{R}^{2n}$ is the concatenation of (f) and (g).

-$\boldsymbol{C}_n(f)$ is the n-dimensional anticirculant matrix of f .When it is clear from context, we will drop the subscript n, and just write $\boldsymbol{C}(f)$.

2. Lattices

An n-dimensional lattice is a full-rank discrete subgroup of $\mathbb{R}^n$. Here we focus on NTRU lattice.

Definition 1 (NTRU lattice) Let q be the prime bigger than 5 and n be a power of 2. And polynomials f, $g \in R_q$(f is invertible modulo q). Let $h = g * f^{-1} \bmod q$. The NTRU lattice associated to h and q is $\Lambda_{h,q} = \{(u, v) \in R^2 | u + v * h = 0 \bmod q\}$.

3. Gaussian on lattices

Gaussian sampling was first proposed in Ref.[12] as a technique to use a short basis as a trapdoor without leaking any information about the short basis. The discrete Gaussian distribution on lattice is defined as follows.

Definition 2 (Discrete Gaussians distribution) For any $s > 0$, $c \in \mathbb{R}^n$, define the n-dimensional Gaussian function $\rho_{s,c} : \mathbb{R}^n \to (0, 1]$ as

$$\rho_{s,c}(x) \triangleq \exp(-\pi \frac{||x-c||^2}{s^2}).$$

For any lattice Λ, $\rho_{s,c}(\Lambda) \triangleq \sum_{x \in \Lambda} \rho_{s,c}(x)$. The probability mass function of the discrete Gaussian distribution is normalized as $D_{\Lambda,s,c}(x) = \rho_{s,c}(x)/\rho_{s,c}(\Lambda)$. For simplicity, in the rest of the paper, $D_{\Lambda,s,c}(x)$ will be abbreviated as $D_{\Lambda,s}(x)$.

In the following lemmas, we review several well-known facts about discrete Gaussian distribution.

Lemma 1 (refer to Ref.[29]) For any n-dimensional lattice Λ, center $c \in \mathbb{R}^n$,the smoothing parameter $\eta_\varepsilon(\Lambda)$, positive $\varepsilon > 0$, $s > 2\eta_\varepsilon(\Lambda)$. For any $x \in \Lambda$, we have

$$D_{\Lambda,s,c}(x) \le (\frac{1+\varepsilon}{1-\epsilon})2^{-n}$$

Lemma 2 For any $\sigma > 0$ and positive integer m, we have

1) $\Pr[x \longleftarrow D_{\mathbb{Z},\sigma} : |x| > 12\sigma] < 2^{-100}$.

2) $\Pr[x \longleftarrow D_{\mathbb{Z}^m,\sigma} : ||x|| > \sigma\sqrt{m}] < 2^{-m}$.

Lemma 3 (refer to Ref.[25]) For any $v \in \mathbb{Z}^m$ and a positive real α, if $\sigma = \Omega(||v||\sqrt{\log m})$, we have

$\Pr[x \leftarrow D_{\mathbb{Z}^m,\sigma} : D_{\mathbb{Z}^m,\sigma}(x)/D_{\mathbb{Z}^m,\sigma,v}(x) = O(1)]$
$=1 - 2^{-\Omega(\log m)}$.

4. Hardness assumption

Definition 3 (R-SIS$_{q,2,\beta}$) A way to state the SIS problem on the NTRU lattice $\Lambda_{h,q}$ is to set $R = \mathbb{Z}[x]/(x^n+1)$, small polynomials f, g are picked according to Fig.3 in Ref.[16], and $h = g/f$. So R-SIS$_{q,2,\beta}$ is to find the (z_1, z_2) which satisfies $(z_1, z_2) \in \Lambda_{h,q}$ and $||(z_1, z_2)|| \le \beta$.

Theorem 1 (Theorem2.1 of Ref.[30]) Let $n = 2^t, \Phi = x^n + 1$ and $\varepsilon > 0$. And m, q are positive integers such that $q \ge \beta\sqrt{n} \cdot \Omega(\log n)$ and $m, \log q \le Poly(n)$. If there exists a polynomial-time algorithm $\mathcal{A}$ solving R-SIS$_{q,m,\beta}$ with non-negligibility probability, a new algorithm $\mathcal{B}$ can be constructed to solve γ-Ideal-Shortest vector problem (γ-Ideal-SVP) in polynomial-time with $\gamma \ge \beta\sqrt{n} \cdot \Omega(\log n)$ by invoking algorithm $\mathcal{A}$.

Just let $m = 2$, according to Theorem 1, the R-SIS$_{q,2,\beta}$ problem can be reduced to the γ-Ideal-SVP.

III. Syntax and Security Model for CLSAS Schemes

1. Syntax

A CLSAS scheme is 7-tuplels of algorithms: *Setup*, *Extract-Partial-Private-Key*, *Set-Secret-Value*,

Set-Private-Key, *Set-Public-Key*, *CL-AggSign*, *CL-AggVerify*.

Setup(n). Taking the security parameter n as input, PKG outputs the master private/public key pair (msk, mpk). Note that PKG keeps msk secret.

Extract-Partial-Private-Key(id_i). On input of the identity id_i of user U_i, PKG outputs a partial private key d_i to the user. Here $1 \le i \le k$.

Set-Secret-Value(id_i). Given the identity id_i, the user U_i outputs a secret value s_i.

Set-Private-Key(d_i, s_i). Taking d_i and s_i as input, the user U_i outputs sk_i as the full private key.

Set-Public-Key(sk_i). On input of full private key sk_i, the user U_i outputs a public key pk_i.

CL-AggSign$(sig_{i-1}, \overrightarrow{pk}_{i-1}, mpk, \overrightarrow{m}_{i-1}, \overrightarrow{id}_{i-1}, \overrightarrow{sig}_{i-1}, m_i, id_i)$. Given the current user's identity id_i, the current message m_i, the previous message set $\overrightarrow{m}_{i-1}$, the previous identity set $\overrightarrow{id}_{i-1}$, the previous signature set $\overrightarrow{sig}_{i-1}$,the previous public key set $\overrightarrow{pk}_{i-1}$, an aggregate signature sig_{i-1} of U_{i-1}, U_i firstly checks whether *CL-AggVerify*$(sig_{i-1}, \overrightarrow{pk}_{i-1}, mpk, \overrightarrow{m}_{i-1}, \overrightarrow{id}_{i-1}, \overrightarrow{sig}_{i-1})$=1, if so, U_i continues to compute the signature sig_i. Finally, U_i adds $sig_i, m_i, id_i, pk_i$to the corresponding sets.

CL-AggVerify$(sig_i, \overrightarrow{pk}_i, mpk, \overrightarrow{m}_i, \overrightarrow{id}_i, \overrightarrow{sig}_i)$. On input of sig_i, $\overrightarrow{pk}_i$, mpk, $\overrightarrow{m}_i$, $\overrightarrow{id}_i$ and $\overrightarrow{sig}_i$, the verifier checks whether $\overrightarrow{sig}_i$ is a valid certificateless aggregate signature generated by the user U_i on messages $\overrightarrow{m}_i$, if so, outputs 1. Otherwise, outputs 0.

2. Security model for CLSAS scheme

In general, a secure CLSAS scheme should satisfy the following requirements:

1) Correctness: The certificateless sequential aggregate signature obtained from CL-AggSign can be verified by the verifier.

2) Unforgeability: When it comes to the unforgeability of the CLSAS scheme, we should consider two types of adversaries.

Type 1 The adversary $\mathcal{A}_1$ models an outside attacker. So the Type 1 adversary $\mathcal{A}_1$ can replace any user's public key with the value chosen by himself.

Type 2 The adversary $\mathcal{A}_2$ models a malicious PKG. So the Type 2 adversary $\mathcal{A}_2$ knows the master secret key msk.

However, neither Type 1 adversary nor Type 2 adversary can replace public keys and know the master secret key at the same time.

IV. A CLSAS Scheme from NTRU Lattices

1. Construction

Let prime $q = \widetilde{\Omega}(\beta\sqrt{n}) \ge 2, n$ be the security parameter, κ, k, Λ be the positive integers, $s = \Omega(\sqrt{n \log q}), \sigma = 12s\kappa n, H : \{0,1\}^* \to \{\boldsymbol{v} \in \mathbb{Z}_q^n\}$ and $H_1 : \mathbb{Z}_q^{2n} \times \{0,1\}^* \to D_{H1} : \{\boldsymbol{v} \in \mathbb{Z}_q^n, 0 \le ||\boldsymbol{v}||_1 \le \kappa, \kappa \ll q\}, H_{11}, H_{12}, H'_{11}, H'_{12} : \{0,1\}^* \to \{0,1\}^l, \mathrm{f}_h : R''^2 \to R_q, G_{\mathrm{f}_h} : \{0,1\}^l \to R'$ and enc:$\{0,1\}^* \to \{0,1\}^* \times R'$. Where k is the number of signers, $R' = \{\boldsymbol{v} \in R_q, ||\boldsymbol{v}|| \le \sigma\sqrt{n}/2\}, \log(R') > l, l < n$ and $R'' = \{\boldsymbol{v} \in R_q, ||\boldsymbol{v}|| \le \sigma\sqrt{n}\}$. Our certificateless sequential aggregate signature scheme on NTRU lattice is:

Setup(n). Taking the security parameter n as input, the PKG runs the trapdoor generating algorithm to output a trapdoor $\boldsymbol{B} = \begin{pmatrix} \boldsymbol{C}(f) & \boldsymbol{C}(g) \\ \boldsymbol{C}(F) & \boldsymbol{C}(G) \end{pmatrix}$ as well as $h \in R_q^\times$. And ensuring $(1+h) \notin R_q^\times$ holds. Where (F,G) is the output of Babai's nearest plane algorithm to approximate pair (F_q, G_q) by an integer linear combination of (f, g). Here F_1, G_1, F_q and G_q satisfy $fG_1 - gF_1 = 1$,$F_q = qF_1$, $G_q = qG_1$. Then, PKG outputs $\boldsymbol{B}$ and h as the msk and mpk, respectively.

Extract-Partial-Private $-$ *Key*(msk, id_i). Taking the master private key msk and identity id_i as input, the PKG runs the pre-image sampling algorithm on the NTRU lattice to output $(s_{i,1}, s_{i,2})$. Then the PKG sends $(s_{i,1}, s_{i,2})$ to the user.

Set-Secret-Value(id_i). The user chooses $s'_{i,1}, s'_{i,2} \in D_{\mathbb{Z}^n,s}$ and outputs $s_i = (s'_{i,1}, s'_{i,2})$.

Set-Private-Key(d_i, s_i). Given the user U_i's partial private key d_i and the secret value s_i, the user U_i outputs $sk_i = (d_i, s_i)$ as the full private key.

Set-Public-Key(sk_i). Taking full private key sk_i as input, the user computes $pk_i = s'_{i,1} + s'_{i,2} * h$ and outputs pk_i as the public key.

CL-AggSign$(\Sigma_{i-1}, m_i, sk_i)$. The detailed *CL-AggSign* is shown in Algorithm 1.

CL-AggVerify(Σ_k). The detailed *CL-AggVerify* is shown in Algorithm 2.

2. Correctness

Theorem 2 The lattice-based CLSAS scheme satisfies correctness.

Proof According to the *CL-AggSign* phase, we know

$$\begin{aligned} &\begin{bmatrix} \mathrm{f}_h(sig_{i,1}) \\ \mathrm{f}_h(sig'_{i,1}) \end{bmatrix} - \begin{bmatrix} H(id_i) \\ pk_i \end{bmatrix} * e_i \\ &= \begin{bmatrix} z_{i,1} + z_{i,2} * h \\ z'_{i,1} + z'_{i,2} * h \end{bmatrix} - \begin{bmatrix} H(id_i) \\ pk_i \end{bmatrix} * e_i \\ &= \begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,1} + y'_{i,2} * h \end{bmatrix} \end{aligned}$$

So we get $(y_{i,1} - g_{i,1}) + (y_{i,2} - g_{i,2}) * h = \beta_i(1 + h)$.

Hence,$\beta_i = [\mathrm{f}_h(\sigma_{i,1}) - H(id_i) * e_i - (g_{i,1} + g_{i,2} * h)]/(1+h)$, the verification can be executed successfully. It means the signature satisfies the correctness.

Algorithm 1 *CL-AggSign*

Input:the message m_i, the user's private key sk_i
Output:the signature Σ_i

1: If $i = 1$ then
2: $\Sigma_0 \leftarrow (\varepsilon, \varepsilon, \varepsilon, 0^n)$
3: End
4: Parse Σ_i as $(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1}, e_{i-1}, h_{i-1,1}, h_{i-1,2}, h'_{i-1,1}, h'_{i-1,2})$, where f_h is the trapdoor function.
5: If $CL\text{-}AggVerify(\Sigma_{i-1}) == (\perp, \perp)$ then
6: End
7: Compute $(\alpha_i, \beta_i) \leftarrow \mathrm{enc}_{\mathrm{f}_h}(sig_{i-1})$,
8: Set $\overrightarrow{\alpha}_i = \overrightarrow{\alpha}_{i-1} | \alpha_i$,
9: Compute
$h_{i,1} = h_{i-1,1} \oplus H_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h_{i,2} = h_{i-1,2} \oplus H_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h'_{i,1} = h'_{i-1,1} \oplus H'_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h'_{i,2} = h'_{i-1,2} \oplus H'_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
10: Compute
$g_{i,1} \leftarrow G_{\mathrm{f}_h}(h_{i,1}), g_{i,2} \leftarrow G_{\mathrm{f}_h}(h_{i,2})$
$g'_{i,1} \leftarrow G_{\mathrm{f}_h}(h'_{i,1}), g'_{i,2} \leftarrow G_{\mathrm{f}_h}(h'_{i,2})$
11: Compute
$$y_i = \begin{bmatrix} y_{i,1} = g_{i,1} + \beta_i \\ y_{i,2} = g_{i,2} + \beta_i \end{bmatrix}, y'_i = \begin{bmatrix} y'_{i,1} = g'_{i,1} + \beta_i \\ y'_{i,2} = g'_{i,2} + \beta_i \end{bmatrix}.$$
12: Compute $e_i = H_1(\begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,2} + y'_{i,2} * h \end{bmatrix}, m_i)$
13: Compute
$$sig_i = \begin{bmatrix} z_{i,1} \\ z_{i,2} \\ z'_{i,1} \\ z'_{i,2} \end{bmatrix} = \begin{bmatrix} s_{i,1} \\ s_{i,2} \\ s'_{i,1} \\ s'_{i,2} \end{bmatrix} * e_i + \begin{bmatrix} y_{i,1} \\ y_{i,2} \\ y'_{i,1} \\ y'_{i,2} \end{bmatrix}$$
14: $\Sigma_i \leftarrow (\mathrm{f}_h, \overrightarrow{m}_i, \overrightarrow{\alpha}_i, sig_i, e_i, h_{i,1}, h_{i,2}, h'_{i,1}, h'_{i,2})$ is returned with probability $min(1, \frac{D_{\mathbb{Z}^n,\sigma}(sig_i)}{MD_{\mathbb{Z}^n,\sigma,s_ie_i}(sig_i)})$. If nothing is outputted, repeat this algorithm.

3. Security

Theorem 3 The CLSAS scheme is proven existentially unforgeable against strong adversaries in random oracle model, under the assumption the γ-Ideal-SVP against polynomial-time algorithm is hard.

Lemma 4 If the $(q, 2, (4\sigma + 2\kappa s)\sqrt{2n})$-SIS on NTRU lattice $\Lambda_{h,q}$ is intractable, the new CLSAS scheme is existentially unforgeable against any polynomial-time strong Type 1 adversary in the random oracle model.

Proof Assuming there is a PPT adversary $\mathcal{A}_1$ who breaks the new CLSAS scheme with non-negligible probability, we can construct a simulator $\mathcal{C}$ to solve the SIS problem on NTRU lattice as follows.

Invocation Being invoked on a random instance of the $(q, 2, \beta)$-SIS problem on NTRU lattice $\Lambda_{h,q}$, the simulator $\mathcal{C}$ is required to return a valid solution.

Supplied: two polynomials $h, t \in R_q^\times$ and NTRU lattice $\Lambda_{h,q}$.

Requested: $(u, v) \in \Lambda_{h,q}$.

Assuming that 1) for the same input, the following oracles are queried only once, 2) before any query $Q_k = (\mathrm{f}_h, \overrightarrow{m}_k, \overrightarrow{\alpha}_{k-1}, sig_{k-1})$ to H_{11}, H_{12}, H'_{11} and H'_{12} Oracles, the sequential forger must have made query id_i to H Oracle, queries $h_{i,1}, h_{i,2}, h'_{i,1}, h'_{i,2}$ to G_{f_h} oracle and queries Q_i to H_{11}, H_{12}, H'_{11} and H'_{12} Oracles for $1 \le i \le k-1$, 3) Before any query (Σ_i, m_i, sk_i) to CL-AggSign-Oracle, the sequential forger must have made the necessary G_{f_h}, H_{11}, H_{12}, H'_{11}, H'_{12} and H_1 Oracles query. And the CL-AggVerify$(\Sigma_{i-1}) = 1$.

Queries $\mathcal{A}_1$ can adaptively query all the oracles shown next:

1) H-Oracle query. The simulator $\mathcal{C}$ keeps a list L_H-list=$\{id_i, d_i, s_{i,1} + s_{i,2} * h\}$. Taking an identity $id_i \in \{0,1\}^*$ as input, $\mathcal{C}$ looks up it in L_H-list. If id_i is found in the L_H-list, the simulator $\mathcal{C}$ returns the $s_{i,1} + s_{i,2} * h$ to $\mathcal{A}_1$. Otherwise, $\mathcal{C}$ picks two polynomials $s_{i,1}, s_{i,2}$ from $D_{\mathbb{Z}^n,s}$, stores $(id_i, d_i = (s_{i,1}, s_{i,2}), s_{i,1}+s_{i,2}*h)$ and returns $s_{i,1} + s_{i,2} * h$ successively to $\mathcal{A}_1$.

2) Create-User-Oracle query. The simulator $\mathcal{C}$ keeps a list L_C-list which is 5-tuples $(id_i, d_i, s_i, sk_i, pk_i)$. On input of id_i, $\mathcal{C}$ does as follows. If id_i is found in the L_C-list, then $\mathcal{C}$ returns pk_i. Otherwise, the simulator $\mathcal{C}$ arises the H-Oracle query for d_i. Then the algorithm *Set-Secret-Value*, *Set-Private-Key* and the algorithm *Set-Public-Key* will be performed by the simulator $\mathcal{C}$ to output value $s_i = (s'_{i,1}, s'_{i,2})$, and $sk_i = (d_i, s_i)$, respectively. Finally, the tuples $(id_i, d_i, s_i, sk_i, pk_i)$ will be stored in L_C-list and $\mathcal{C}$ returns to the adversary $\mathcal{A}_1$.

3) Extract-Partial-Private-Key-Oracle query. Taking id_i as input, $\mathcal{C}$ searches the L_C-list for s_i.

4) Replace-Public-Key-Oracle query. On input of id_i and a new public key pk'_i, $\mathcal{C}$ looks up the corresponding public key pk_i and replaced it with pk'_i. Finally, this replacement will be recorded by $\mathcal{C}$ later.

5) $H_{11}, H_{12}, H'_{11}, H'_{12}$ Oracles query. On input of $Q_i = (\mathrm{f}_h, \overrightarrow{m}_i, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$, the simulator $\mathcal{C}$ randomly selects $h_{i,1}, h_{i,2}, h'_{i,1}, h'_{i,2} \in \{0,1\}^l$ and then returns $h_{i,1}, h_{i,2}, h'_{i,1}, h'_{i,2}$ to the adversary $\mathcal{A}_1$.

6) G_{f_h}-Oracle query. On input of $h_{i,j}$ or $h'_{i,j}$ for $1 \le i \le k, 1 \le j \le 2$, the simulator $\mathcal{C}$ randomly selects $g_{i,j}$ and $g'_{i,j} \in R'$ then returns $g_{i,j}$ and $g'_{i,j}$ to $\mathcal{A}_1$.

7) H_1 Oracle query. $\{\begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,1} + y'_{i,2} * h \end{bmatrix}, m_i, e_i\}$ is included in L_{H1}-list. Taking $\begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,1} + y'_{i,2} * h \end{bmatrix}, m_i$ as input, $\mathcal{C}$ looks up them in L_{H1}-list. If they are found in it, $\mathcal{C}$ returns the corresponding e_i. Otherwise, $\mathcal{C}$ randomly selects e_i from D_{H1}, and stores $(\begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,1} + y'_{i,2} * h \end{bmatrix}, m_i, e_i)$ in the L_{H1}-list. Finally, the simulator $\mathcal{C}$ returns e_i to $\mathcal{A}_1$.

8) CL-AggSign-Oracle query. On input of $(\Sigma_{i-1}, m_i, id_i)$. The simulator $\mathcal{C}$ first searches the H-Oracle query for the d_i, then the $CL-AggSign$ algorithm

will be run to return a signature Σ_i Note that if pk_i is the user's current public key(that is to say pk_i has not been replaced), then $x_i = \perp$. In this case, $\mathcal{C}$ can run the $CL-AggSign$ algorithm to generate a valid signature.

Forgery Finally, $\mathcal{A}_1$ outputs a valid forgery $\Sigma'_k = (\mathrm{f}_h, \overrightarrow{m}_k, \overrightarrow{\alpha}_k, sig_k)$ with non-negligible probability. During the execution of $CL\text{-}AggVerify(\Sigma'_i)$ containing id'_i, m'_i, sig'_i and e'_i. Where id'_i has never been queried for Creat-User-Oracle query and Extract-Partial-Private-Key-Oracle query, m'_i has never been queried for CL-AggSign-Oracle.

The simulator $\mathcal{C}$ can solve the SIS problem on NTRU lattice as follows.

After receiving the forgery signature (sig'_i, e'_i) on (m_i, Σ'_{i-1}), $\mathcal{C}$ will output a new forgery (sig^*_i, e^*_i) on the same (m'_i, Σ'_{i-1}) by the forking lemma. So we get $\begin{bmatrix} (z^*_{i,1} - z'_{i,1}) + (z^*_{i,2} - z'_{i,2}) * h \\ (z'^*_{i,1} - z''_{i,1}) + (z'^*_{i,2} - z''_{i,2}) * h \end{bmatrix} = \begin{bmatrix} s^*_{i,1} + s^*_{i,2} * h \\ s'^*_{i,1} + s'^*_{i,2} * h \end{bmatrix} * (e^*_i - e'_i)$. And then the inequality $[(z^*_{i,1} - z'_{i,1}) - s^*_{i,1}(e^*_i - e'_i)] + [(z^*_{i,2} - z'_{i,2}) - s^*_{i,2}(e^*_i - e'_i)] * h = 0$ also holds. Applying Lemma 3, we know the distribution of $z^*_{i,1}, z'_{i,1}, z^*_{i,2}, z'_{i,2}$ is very close to $D_{\mathbb{Z}^n,\sigma}$ with overwhelming probability. Therefore, by Lemma 2, we have $||z^*_{i,1}||, ||z'_{i,1}||, ||z^*_{i,2}||, ||z'_{i,2}|| \leq 2\sigma\sqrt{n}$. So the inequality $||(z^*_{i,1} - z'_{i,1}) - s^*_{i,1} * (e^*_i - e'_i)|| \leq ||z^*_{i,1}|| + ||z'_{i,1}|| + ||s^*_{i,1}|| \cdot ||e^*_i - e'_i|| \leq (4\sigma + 2\kappa s)\sqrt{n}$ holds and $||(z^*_{i,2} - z'_{i,2}) - s^*_{i,2} * (e^*_i - e'_i)|| \leq ||z^*_{i,2}|| + ||z'_{i,2}|| + ||s^*_{i,2}|| \cdot ||e^*_i - e'_i|| \leq (4\sigma + 2\kappa s)\sqrt{n}$.

According to the pre-image min-entropy property, we know there is a partial private key $d'_i = (s_{i',1}, s_{i',2})$ satisfying $s_{i',1} + s_{i',2} * h = H(id'_i)$ and $(s_{i',1}, s_{i',2}) \neq (s^*_{i,1}, s^*_{i,2})$. 1) When $s_{i',1} \neq s^*_{i,1}$, $[(z^*_{i,1} - z'_{i,1}) - s^*_{i,1}(e^*_i - e'_i)] - [(z^*_{i,1} - z'_{i,1}) - s^*_{i',1}(e^*_i - e'_i)] = (s_{i',1} - s^*_{i,1})(e'_i - e^*_i) \neq 0$. Thus, if $[(z^*_{i,1} - z'_{i,1}) - s^*_{i',1}(e^*_i - e'_i)] = 0$, $[(z^*_{i,1} - z'_{i,1}) - s^*_{i,1}(e^*_i - e'_i)] \neq 0$. 2) When $s_{i',2} \neq s^*_{i,2}$, we can also know if $[(z^*_{i,2} - z'_{i,2}) - s^*_{i',2}(e^*_i - e'_i)] = 0$, $[(z^*_{i,2} - z'_{i,2}) - s^*_{i,2}(e^*_i - e'_i)] \neq 0$. According to 1) and 2), we know $([(z^*_{i,1} - z'_{i,1}) - s^*_{i,1}(e^*_i - e'_i)], [(z^*_{i,2} - z'_{i,2}) - s^*_{i,2}(e^*_i - e'_i)]) \neq 0$ holds with probability more than 0.75.

So $([(z^*_{i,1} - z'_{i,1}) - s^*_{i,1}(e^*_i - e'_i)], [(z^*_{i,2} - z'_{i,2}) - s^*_{i,2}(e^*_i - e'_i)])$ is the solution to the SIS problem on NTRU lattice above, where $\beta \geq (4\sigma + 2\kappa s)\sqrt{2n}$.

Lemma 5 If the $(q, 2, (4\sigma + 2\kappa s)\sqrt{2n})$ SIS on NTRU $\Lambda_{h,q}$ is instractable, the new CLSAS scheme is existentially unforgeable against any polynomial-time strong Type 2 adversary in the random oracle model.

Proof the proof is similar to that in Lemma 4, and refer to the full version for it.

According to Lemma 4 and Lemma 5, Theorem 3 holds.

Algorithm 2 $CL\text{-}AggVerify$

Input:Σ_k
Output:
1: Parse Σ_k as $(\mathrm{f}_h, \overrightarrow{m}_k, \overrightarrow{\alpha}_k, sig_k, e_k, h_{k,1}, h_{k,2}, h'_{k,1}, h'_{k,2})$.
2: For $i = k \to 1$, do
3: if $log_2(R') \leq l$ or $sig_{i,1}, sig'_{i,1} \notin R''^2$, then
4: return $(\perp, \perp)$.
5: End
6: Compute
$g_{i,1} \leftarrow G_{\mathrm{f}_h}(h_{i,1}), g_{i,2} \leftarrow G_{\mathrm{f}_h}(h_{i,2})$
$g'_{i,1} \leftarrow G_{\mathrm{f}_h}(h'_{i,1}), g'_{i,2} \leftarrow G_{f_h}(h'_{i,2})$
7:Compute
$\beta_i = [\mathrm{f}_h(sig_{i,1}) - H(id_i) * e_i - (g_{i,1} + g_{i,2} * h)]/(1+h)$.
8: Compute $sig_{i-1} \leftarrow dec_{\mathrm{f}_h}(\alpha_i, \beta_i)$.
9: Compute
$h_{i-1,1} = h_{i,1} \oplus H_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h_{i-1,2} = h_{i,2} \oplus H_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h'_{i-1,1} = h'_{i,1} \oplus H'_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
$h'_{i-1,2} = h'_{i,2} \oplus H'_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$
10: End
11: If $\Sigma_0 = (\varepsilon, \varepsilon, \varepsilon, 0^n)$ then
12: return $(\mathrm{f}_h, \overrightarrow{m}_k)$.
13: End
14: Else return $(\perp, \perp)$.

4. Analysis

There have been two CLS schemes in Ref.[31,32] from lattice which is proven security in random oracle model. And we know the scheme in Ref.[32] is more efficient. Now we compare our new CLSAS scheme with Ref.[32] in Table 1.

Table 1. The efficiency comparison between two schemes

Schemes	Signers' number	Signature size
Ref.[32]	1	$4n\log 2\sigma + \kappa$
This work	k	$4n\log(2\sigma) + nk(4\log 2\sigma - \log q) + \kappa + 4l$

From Table 1, we know the compression rate of our new scheme is $rate(k)$, where

$$\begin{aligned} rate(k) =& 1 - \frac{4n\log(2\sigma) + nk(4\log 2\sigma - \log q) + \kappa + 4l}{k(4n\log 2\sigma + \kappa)} \\ \geq& 1 - (\frac{1}{k} + \frac{4n(\log 2\sigma + \kappa - \log q^{1/4}) + 4l/k}{4n\log 2\sigma + \kappa}) \\ \geq& 1 - (\frac{1}{k} + \frac{o(\log q)}{o(\log q + \log(q^{1/4})}) \end{aligned}$$

For $q = n^{2c}$, $rate(k) \approx 1 - 1/k - 1/c$. When c is large enough, the scheme is almost optimal. That is to say, the signature's size in our new scheme is $1/k$ of the individual signatures' size.

V. An Extensional CBSAS Scheme from NTRU Lattice

1. Construction

The extensional CBSAS scheme works as follows:

$Setup(n)$. Taking the security parameter n as input,

the PKG runs the trapdoor generating algorithm to output a trapdoor $\boldsymbol{B} = \begin{pmatrix} \boldsymbol{C}(f) & \boldsymbol{C}(g) \\ \boldsymbol{C}(F) & \boldsymbol{C}(G) \end{pmatrix}$ as well as $h \in R_q^{\times}$. And ensuring that $(1+h) \notin R_q^{\times}$. Finally, PKG outputs $\boldsymbol{B}$ and h as the msk and mpk, respectively.

$Set\text{-}User\text{-}Key(id_i)$. Given the identity id_i, the user chooses $s'_{i,1}, s'_{i,2} \in D_{\mathbb{Z}^n,s}$ and then sets his public/private key pair as $pk_i = s'_{i,1} + s'_{i,2} * h, sk_i = (s'_{i,1}, s'_{i,2})$.

$Extract\text{-}Certificate(msk, id_i)$. Taking the master private key msk and identity id_i as input, the PKG runs the pre-image sampling algorithm on the NTRU lattice to output $(s_{i,1}, s_{i,2})$ to the user as the certificte c_i.

$CB\text{-}AggSign(\Sigma_{i-1}, m_i, sk_i)$. The $CB\text{-}AggSign$ algorithm is shown in Algorithm 3.

$CB\text{-}AggVerify(\Sigma_k)$. The $CB\text{-}AggVerify$ is the same as $CL\text{-}AggVerify$ in Section IV.

Algorithm 3 *CB-AggSign*

Input:the message m_i, the user's private key sk_i

Output:the signature Σ_i

1: If $i = 1$ then

2: $\Sigma_0 \leftarrow (\varepsilon, \varepsilon, \varepsilon, 0^n)$

3: End

4: Parse Σ_i as $(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1}, e_{i-1}, h_{i-1,1}, h_{i-1,2}, h'_{i-1,1}, h'_{i-1,2})$, where f_h is the trapdoor function.

5: If $CL\text{-}AggVerify(\Sigma_{i-1}) == (\perp, \perp)$ then

6: End

7: Compute $(\alpha_i, \beta_i) \leftarrow \mathrm{enc}_{\mathrm{f}_h}(sig_{i-1})$.

8: Set $\overrightarrow{\alpha}_i = \overrightarrow{\alpha}_{i-1} | \alpha_i$.

9: Compute

$$h_{i,1} = h_{i-1,1} \oplus H_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$$
$$h_{i,2} = h_{i-1,2} \oplus H_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$$
$$h'_{i,1} = h'_{i-1,1} \oplus H'_{11}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$$
$$h'_{i,2} = h'_{i-1,2} \oplus H'_{12}(\mathrm{f}_h, \overrightarrow{m}_{i-1}, \overrightarrow{\alpha}_{i-1}, sig_{i-1})$$

10: Compute

$$g_{i,1} \leftarrow G_{\mathrm{f}_h}(h_{i,1}), g_{i,2} \leftarrow G_{\mathrm{f}_h}(h_{i,2})$$
$$g'_{i,1} \leftarrow G_{\mathrm{f}_h}(h'_{i,1}), g'_{i,2} \leftarrow G_{\mathrm{f}_h}(h'_{i,2})$$

11: Compute

$$y_i = \begin{bmatrix} y_{i,1} = g_{i,1} + \beta_i \\ y_{i,2} = g_{i,2} + \beta_i \end{bmatrix}, y'_i = \begin{bmatrix} y'_{i,1} = g'_{i,1} + \beta_i \\ y'_{i,2} = g'_{i,2} + \beta_i \end{bmatrix}$$

12: Compute $e_i = H_1(\begin{bmatrix} y_{i,1} + y_{i,2} * h \\ y'_{i,2} + y'_{i,2} * h \end{bmatrix}, m_i)$.

13: Compute

$$sig_i = \begin{bmatrix} z_{i,1} \\ z_{i,2} \\ z'_{i,1} \\ z'_{i,2} \end{bmatrix} = \begin{bmatrix} s_{i,1} \\ s_{i,2} \\ s'_{i,1} \\ s'_{i,2} \end{bmatrix} * e_i + \begin{bmatrix} y_{i,1} \\ y_{i,2} \\ y'_{i,1} \\ y'_{i,2} \end{bmatrix}$$

14: $\Sigma_i \leftarrow (\mathrm{f}_h, \overrightarrow{m}_i, \overrightarrow{\alpha}_i, sig_i, e_i, h_{i,1}, h_{i,2}, h'_{i,1}, h'_{i,2})$ is returned with probability $min(1, \dfrac{D_{\mathbb{Z}^n,\sigma}(sig_i)}{MD_{\mathbb{Z}^n,\sigma,s_ie_i}(sig_i)})$. If nothing is outputted, repeat this algorithm.

The correctness proof is similar to that in Section IV.

2. Security

Theorem 4 The extensional CBSAS scheme is proven existentially unforgeable against strong adversaries in the random oracle model, under the assumption the γ-Ideal-SVP against polynomial-time algorithm is hard.

Proof As can be seen in Section V.1, the extensional CBSAS scheme can be derived from the CLSAS scheme in Section IV.1. And we can directly obtain the security proof from Theorem 3 and the result of Ref.[33]. In detail, the extensional CBSAS scheme is also secure against strong adversaries in the random oracle model, under the assumption the γ-Ideal-SVP against polynomial-time algorithm is hard.

VI. Conclusions

With the significant advantages of certificateless, no escrow feature, low computations and communications cost, the CLSAS scheme has absorbed the general attention since it was proposed. And lattice has been shown the best candidate in quantum era. In this paper, we use the trapdoor sampling technique on NTRU lattice to construct a collision-resistant trapdoor function, based on this trapdoor function, the first CLSAS scheme on NTRU lattice is proposed. And its security can be reduced to the γ-Ideal-SVP on lattice. So it is proved secure in random oracle model and is still secure even in quantum era. Then we extended the CLSAS scheme into a CBSAS scheme on NTRU lattice, which is also secure in quantum era. An efficient lattice-based CLSAS and CBSAS schemes in standard model will be our future work.

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